

Ekaterina Landgren

Contact Information

Department of Environmental Social Sciences
Stanford Doerr School of Sustainability
Stanford University
Stanford, California, 94305

kath.landgren@stanford.edu
kathlandgren.com

Research interests

Complex social systems: environmental social science, computational social science, opinion dynamics
Mathematics of climate: conceptual climate models, exoplanetary atmosphere dynamics

Education

Cornell University 2022

Ph.D., Applied Mathematics

“Models of Varying Complexity from Voter Networks to Extrasolar Planets”

Advisor: Steven H. Strogatz

Cornell University 2020

M.S., Applied Mathematics

Brown University 2017

B.S., Applied Mathematics, B.A., Philosophy

Cum Laude, Phi Beta Kappa, Sigma Xi

“Modeling Evacuation Dynamics in a Crowded Room”

Advisor: Bjorn Sandstede

Professional Experience

Stanford University September 2025–present

Stanford Doerr School of Sustainability

Dean’s Sustainability Leaders Postdoctoral Fellow

University of Colorado Boulder

Cooperative Institute for Research in Environmental Sciences

Postdoctoral Researcher 2025

Visiting Postdoctoral Fellow 2023–2024

Awards and Fellowships

Best Postdoc Presentation 2026

Awarded at the Sustainability Data Science Conference held at Stanford University.

Rising Star in Computational and Data Sciences 2025

Oden Institute, University of Texas at Austin

Best Oral Presentation by an Early-Career Researcher, Honorable Mention 2024

Awarded at International School and Conference on Network Science (NetSci 2024).

SIAM Science Policy Fellowship	2024
Awarded annually to 5 early-career mathematicians to gain in-depth knowledge of science policy.	
Collaborate@ICERM	2024
Awarded to a team of 5 mathematicians to spend a week collaborating on the project “Modeling and Analysis of Candidate Momentum in U.S. Primary Elections.”	
Zonta International Amelia Earhart Fellowship	2021
Awarded annually to up to 35 women around the globe pursuing a PhD in space sciences.	
SIAM Student Chapter Certificate of Recognition	2021
Awarded for outstanding service and contributions to the SIAM student chapter.	
Undergraduate Research and Teaching Award	2015, 2016
Awarded to Brown students collaborating with faculty on research projects.	
Mathematical Contest in Modeling, Honorable Mention	2016
In an undergraduate team, created, analyzed, and wrote a report on a model of fluid dynamics.	
Brown Mathematical Contest for Modeling, Outstanding Winner	2015
In an undergraduate team, created, analyzed, and wrote a report on a model of Hantavirus spread.	
Travel awards	
yrCSS Bridge Grant	2025
Dynamics Days 2025 Travel Award	2025
AWM Travel Grant	2024
Postdoctoral Association of Colorado Travel Award	2024
yrCSS Scholarship for Events on Complex Systems	2024
SIAM Early Career Travel Award	2024
CIRES Early Career Travel Award	2024
Dynamics Days 2024 Travel Award	2024
SIAM Student Travel Award	2019

Submitted Manuscripts

*Equal contribution indicated by **

1. Consensus and fragmentation in academic publication preferences
Ian Van Buskirk, Marilena Hohmann*, **Ekaterina Landgren***, Johan Ugander, Aaron Clauset,
Daniel Larremore
arXiv preprint: <https://arxiv.org/abs/2603.00807>

Peer-Reviewed Publications

1. U.S. television news coverage of climate change policy is aggregately balanced but polarized
Ekaterina Landgren, Jeremiah Osborne-Gowey, Joshua Garland, Maxwell Boykoff,
Matthew Burgess
Environmental Research Communications 8 051016 (2026). DOI: [10.1088/2515-7620/ae6b10](https://doi.org/10.1088/2515-7620/ae6b10)

Alphabetical author order indicated by ♦

2. Visualizing epidemiological models for policy: design principles for effective communication
Liza Hadley, Nick Holliman, Kai Xu, Edyta Bogucka, Xinhuan Shu, Daniel Archambault,
Ekaterina Landgren, Ellen M. DeGennaro, Stephen M. Kissler.
Frontiers in Public Health 14:1798154 (2026). DOI: [10.1103/dfw8-vhjk](https://doi.org/10.1103/dfw8-vhjk)
Preprint: <https://doi.org/10.17605/OSF.IO/C93HK>
3. Optimal ambition in business, politics, and life
Ekaterina Landgren, Ryan Langendorf, Matthew Burgess
In press at *Physical Review E* (2026). DOI: [10.1103/dfw8-vhjk](https://doi.org/10.1103/dfw8-vhjk)
arXiv preprint: <https://arxiv.org/abs/2502.10500>
4. A Shallow-water Model Exploration of Atmospheric Circulation on Sub-Neptunes: Effects of
Radiative Forcing and Rotation Period
Ekaterina Landgren, Alice Nadeau, Nikole Lewis, Tiffany Kataria, Peter Hitchcock
Planetary Science Journal, 4(6), 106. (2023). DOI: [10.3847/PSJ/acd551](https://doi.org/10.3847/PSJ/acd551)
5. SWAMPE: A Shallow-Water Atmospheric Model in Python for Exoplanets
Ekaterina Landgren, Alice Nadeau
Journal of Open Source Software 7 (80), 4872 (2022). DOI: [10.21105/joss.04872](https://doi.org/10.21105/joss.04872)
6. Comparison of Two Analytic Energy Balance Models Shows Stable Partial Ice Cover Possible for
Any Obliquity
Ekaterina Landgren, Alice Nadeau
Planetary Science Journal 3.79 (2022). DOI: [10.3847/PSJ/ac603d](https://doi.org/10.3847/PSJ/ac603d)
7. How a minority can win: Unrepresentative outcomes in a simple model of voter turnout
Ekaterina Landgren, Jonas L. Juul, Steven H. Strogatz
Physical Review E 104.5 (2021): 054307. DOI: [10.1103/PhysRevE.104.054307](https://doi.org/10.1103/PhysRevE.104.054307)
8. Fractal Behavior of the Fibonomial Triangle Modulo Prime p , Where the Rank of Apparition of p
is $p + 1$
♦ Michael DeBellevue, **Ekaterina Kryuchkova (Landgren)**
Fibonacci Quarterly 56 (2018): 113-120.

Other writing

1. Modeling Social Systems: Transparency, Reproducibility, and Responsibility
♦ Maximino Aldana, Heather Zinn Brooks, Phil Chodrow, Fillipe Georgiou, Joseph Johnson,
Krešimir Josić, Zachary Kilpatrick, **Ekaterina Landgren**, Andrew Nugent, Maurizio Porfiri, Nancy
Rodriguez, Pablo Suárez-Serrato, Roni Barak Ventura, David White, Alexander Wiedemann, Sam
Zhang.
SIAM News 59.2 (2026). DOI: [10.48550/arXiv.2508.18542](https://doi.org/10.48550/arXiv.2508.18542)

Presentations

Invited presentations

1. *Perceived climate polarization from networks to newsrooms* March 2026
Lund University COSHE Seminar
2. *Publication preferences of academics* October 2025
University of Virginia, Seminar, Ahn lab
3. *Perceived climate polarization from networks to newsrooms* July 2025
IT University of Copenhagen Seminar, Copenhagen, Denmark
4. *Disentangling climate polarization* June 2025
Mathematics of Climate Research Network Colloquium
5. *Disentangling climate polarization* June 2025
BIRS Workshop: Collective Social Phenomena
Casa Matemática Oaxaca, Oaxaca, Mexico
6. *Unpacking Climate Polarization Across News and Social Media* May 2025
SIAM Conference on Applied Dynamical Systems, Denver, CO
7. *Disentangling climate polarization* April 2025
Rising Stars in Computational and Data Sciences, Oden Institute, Austin, TX
8. *Disentangling climate polarization on Reddit* April 2025
CIRES Early-Career Assembly Seminar, Boulder, CO
9. *Modeling misperception of public support for climate policy* November 2024
University of Minnesota, School of Mathematics, Minneapolis, MN
10. *Exoplanets in 1D and 2D: from ice cover to atmospheric circulation* November 2024
University of Minnesota, Mathematics of Climate Seminar, Minneapolis, MN
11. *Modeling misperception of public support for climate policy* October 2024
Stanford University, Climate Cognition and Sustainability and Social Change Lab
12. *Modeling misperception of public support for climate policy* April 2024
National Ecological Observatory Network (NEON) Science Seminar
13. *Modeling misperception of public support for climate policy* March 2024
University of Vermont, Complex Systems Center, Burlington, VT
14. *Modeling misperception of public support for climate policy* February 2024
University of Colorado Boulder, Dynamical Systems Seminar, Boulder, CO
15. *Modeling misperception of public support for climate policy* February 2024
University of Minnesota, Mathematics of Climate Seminar
16. *Modeling misperception of public support for climate policy* December 2023
University of Colorado Boulder, Mathematical Biology Seminar
17. *A Shallow Water Model of Atmospheric Circulation on Sub-Neptunes* November 2023
Max Planck Institute for Astronomy, Exocoffee
18. *Misperception of public support for climate policy: A Networks Perspective* October 2023
University of Cambridge Centre for Climate Repair, Cambridge, UK
19. *Beyond Echo Chambers: Misperception of Public Support for Climate Policy* September 2023
Brown University, LCDS Seminar, Providence, RI

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| 20. <i>Modeling Misperception of Public Support for Climate Policy</i>
SIAM Conference on Applied Dynamical Systems, Portland, OR | May 2023 |
| 21. <i>A Shallow-Water Model Exploration of Atmospheric Circulation on Sub-Neptunes</i>
Southwest Research Institute, Boulder, CO | April 2023 |
| 22. <i>How Can Minority Win?</i>
University of Colorado Boulder, Seminar, Clauset & Larremore lab group | February 2023 |
| 23. <i>Introduction to Research</i>
Cornell Chapter of Association for Women in Mathematics, Ithaca, NY | February 2022 |
| 24. <i>Effects of Network Structure on Undemocratic Outcomes</i>
Clarkson University, Graduate Student Seminar | August 2021 |
| 25. <i>Effects of Network Structure on Undemocratic Outcomes</i>
SIAM Conference on Applied Dynamical Systems | May 2021 |
| 26. <i>Noisy El Niño: A Case Study of Conceptual Climate Models</i>
Mount Holyoke College, Math and Statistics Tea | March 2021 |
| 27. <i>When Can Minority Win? A Simple Model of Voter Turnout</i>
Women in Network Science Seminar, University of Washington | February 2021 |
| 28. <i>Snowball Planets: Effects of Obliquity, Albedo, and Heat Transport on Ice Cover</i>
Jet Propulsion Laboratory, Exoplanet Journal Club | October 2020 |

Contributed presentations

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| ○ <i>Climate attribution in local news coverage of natural hazards</i>
Sustainability Data Science Conference, Stanford University | April 2026 |
| ○ <i>The publication preferences of academics</i>
International Conference on Computational Social Science, Norrköping, Sweden | July 2025 |
| ○ <i>Optimal ambition in business, politics, and life</i>
International Conference on Computational Social Science, Norrköping, Sweden | July 2025 |
| ○ <i>Modeling misperception of public support for climate policy</i>
International School and Conference on Network Science, Québec City, Canada | June 2024 |
| ○ <i>Modeling misperception of public support for climate policy</i>
SIAM Conference on Mathematics of Planet Earth, Portland, OR | June 2024 |
| ○ <i>Modeling misperception of public support for climate policy</i>
Network Inequality Seminar, Complexity Science Hub | April 2024 |
| ○ <i>Modeling misperception of public support for climate policy</i>
Dynamics Days 2024. University of California, Davis | January 2024 |
| ○ <i>Climate policy is more popular than most people think</i>
Social and Environmental Futures Workshop, University of Colorado Boulder | October 2023 |
| ○ <i>How can minority win?</i>
Contagion on Complex Social Systems Workshop, University of Colorado Boulder | August 2022 |
| ○ <i>Introducing SWAMP-E: Shallow Water Atmosphere Model in Python for Exoplanets</i>
Emerging Researchers in Exoplanet Science Conference | May 2021 |

Poster presentations

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| ○ <i>Optimal Ambition in Business, Politics, and Life</i>
Dynamics Days 2025. Denver, CO | January 2025 |
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- *Modeling misperception of public support for climate policy* July 2024
International Conference on Computational Social Science, Philadelphia, PA
- *Climate policy is more popular than you think!* March 2024
STEM Poster Day at the Colorado State Capitol
Project Bridge, University of Colorado Anschutz
- *Exploring the Interaction of Rotation Rate and Stellar Irradiation on Synchronously Rotating Sub-Neptunes* December 2022
American Geophysical Union Fall Meeting. Chicago, IL
- *Introducing SWAMP-E: Shallow-Water Atmospheric Model in Python for Exoplanets* December 2021
American Geophysical Union Fall Meeting
- *Introducing SWAMP-E: Shallow-Water Atmospheric Model in Python for Exoplanets* May 2021
Emerging Researchers in Exoplanet Science Conference

Student Mentorship

Ian van Buskirk 2024–2025

Ph.D. Student in Computer Science at the University of Colorado Boulder
Mentored jointly with Dan Larremore.

Project title: "Publication preferences of academics"

News Transcript Annotation Team 2023–2024

Team of undergraduate and master's students at the University of Colorado Boulder
Callie Blaseg, Kai Elznic, Lucas Gauthier, Katherine Konecny, Piper Nilsen, Vraj Patel, Trishala Thakur
Led twice-weekly intercoder reliability meetings and supervised news transcript annotations.
Mentored jointly with Jeremiah Osborne-Gowey.

Project title: "Misperception of public support for climate policy"

Ashley Dancer 2023

Ph.D. Student in Environmental Studies at the University of Colorado Boulder
Mentored jointly with Matt Burgess.

Project title: "Agent-Based Model of Fertility"

Thomas Mitchell 2022

Undergraduate Student in Astronomy at Cornell University
Mentored jointly with Nikole Lewis.

Project title: "Energy Balance Model for HAT-P-2b"

Anna Asch 2021

Undergraduate Student in Mathematics at Cornell University
Mentored jointly with Shriya Nagpal and Alice Nadeau.

Project title: "Wind farm layout optimization"

Anna Asch 2020

Undergraduate Student in Mathematics at Cornell University
Directed Reading Program

Project title: "Mathematics and Climate"

Anushka Naranayan 2020

Undergraduate Student in Mathematics at Cornell University

Mentored jointly with Alice Nadeau.

Project title: "Applying the Budyko Model to Martian Obliquity"

Teaching Experience

MIT Educational Studies Program

Instructor

M14095: Mathematical Models and How to Build One, *Online* Summer 2020

Designed and taught a six-session class in mathematical modeling for high school students.

Cornell University

Teaching Assistant

MATH 4210: Nonlinear Dynamics and Chaos Spring 2020

MATH 3610: Mathematical Modeling Fall 2019

MATH 2930: Differential Equations for Engineers Spring 2019

Brown University

Teaching Assistant

APMA 1650: Statistical Inference I Fall 2015, Spring 2017

Industry experience

IMA Math-to-Industry Bootcamp III Summer 2018

Six-week coding and research program. Minneapolis, MN

Hewlett-Packard Customer Operations Summer 2014

Summer intern. Moscow, Russia

Service and Leadership

Conference Session Organizer

- *Human behavior in climate opinion, policy, and environmental systems at SIAM DS25* May 2025

Co-organizer

- *AMS Special Session on Complex Social Systems at JMM* January 2024

Co-organizer

- *Dynamics of Influence and Representation in Social Systems at SIAM DS21* May 2021

Co-organizer

Stanford University

- *Menlo School Catalyst Leadership Program* March 2026

Guest speaker on career pathways in environmental social science

- *Women in Network Science Society*

Treasurer

2025–present

Communications team: publish a monthly newsletter.

2023–present

University of Colorado Boulder

- *CIRES Cross-Discipline and Community Building Mentoring Workshop* April 2025

Panelist

2023–present

- *Postdoctoral Mentoring Program* 2024
Mentor
- *Mathematics of Climate Research Network Mentoring Program* 2024
Mentor
- *Kent Denver School Gender Advancements in STEM Career Panel* January 2023
Panelist

Cornell University

- *Expanding Your Horizons Conference* 2021
Logistics chair, organized a campus-wide STEM outreach event for 500 middle-school girls.
- *Write a Researcher* 2021
Corresponded with a high school student about mathematics research.
- *Center for Applied Mathematics First-Year Mentoring Program* 2019, 2021
Mentored a first-year PhD student.
- *SIAM Graduate Student Chapter* 2018–2021
President. Organized SIAM-sponsored events for student chapter members.
- *Center for Applied Math Anti-Racism Reading Group* 2020
Co-organizer. Moderated a biweekly graduate student discussion focusing on DEI topics.
- *ZigZag Mentorship Program* 2017, 2019
Mentored undergraduate students on course selection and career development.

Brown University

- *Applied Mathematics Department Undergraduate Group* 2015, 2016
President. Organized events for undergraduates interested in applied mathematics.
- *Technology House* 2016
President. Led a sixty-person, communal living group for students interested in STEM topics.
- *New Scientist Program* 2015
Mentored and advised a first generation college student.

Reviewer for

Journal of Open Source Software, npj Complexity, Scientific Reports, Europhysics Letters, Physica D: Nonlinear Phenomena, IC2S2 2025, IC2S2 2026, Climatic Change, Environmental Research: Climate

Other Professional Activities

Workshops attended

- *Robust Institutional Design in Expert Decision Maker Systems*, Santa Fe Institute, NM April 2026
- *Complex Networks Winter Workshop*, Québec City, Canada December 2025
- *Social and Environmental Futures Workshop*, Boulder, CO October 2023
- *Mathematics Research Communities: Complex Social Systems*, Buffalo, NY June 2023
- *Contagion on Complex Social Systems*, Boulder, CO August 2022
- *Science Communication Workshop*, Ithaca, NY October 2021
- *Code/Astro: A Software Engineering Workshop for Astronomy*, Online June 2021

Pedagogical development

- *Research Mentor Training* Spring 2022

Center for the Integration of Research, Teaching and Learning Network
Completed a six-week online course to improve research mentoring skills

Membership in professional organizations

- Society for Industrial and Applied Mathematics
- American Mathematical Society
- Network Science Society
- Mathematics of Climate Research Network
- Women in Network Science Society

Media features

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| ○ Featured in “Postdoctoral fellows build research and community connections” in Stanford Doerr School of Sustainability news (link) | April 2026 |
| ○ Featured in “2024 SIAM Science Policy Fellows” in SIAM news (link) | March 2024 |
| ○ SIAM DS23 presentation featured in SIAM News Blog (link) | May 2023 |
| ○ Amelia Earhart Fellowship Spotlight in the Cornell Chronicle (link) | May 2021 |